

Journal Entry IHJ001: Security Audit Report

Date: 25 th May, 2026	Entry: IHJ001
Description	This entry documents a security and compliance audit conducted for Botium Toys to evaluate current system safeguards and identify critical gaps under NIST CSF, PCI-DSS, and GDPR standards.
Tool(s) used	NIST CSF v1.1 framework, PCI-DSS v4.0 checklist, and GDPR compliance guidelines.
The 5 W's (Who, What, where, when and why)	• Who: Matthew Abegyah (Lead Cybersecurity Analyst) auditing Botium Toys, Inc. • What: The audit revealed critical security gaps, including unencrypted payment card details and customer PII stored in plaintext, universal employee access to internal databases, and a lack of intrusion detection systems and data backups. • Where: The review covered the entire corporate network, on-site offices, warehouse facilities, and the internal database hosting the digital storefront. • When: The assessment took place during the company's expansion into European Union markets. • Why: Rapid business growth occurred without implementing baseline security controls or an asset management program, leaving the company exposed to ransomware attacks, data theft, and severe regulatory fines under PCI-DSS and GDPR
Additional notes	• Immediate remediation requires enforcing AES-256 encryption on all stored cardholder and customer data, restricting database access using least privilege, installing a network intrusion detection system, and creating regular data backups with a formal disaster recovery plan. Existing physical security controls and the 72-hour breach notification procedure are fully compliant.

Journal Entry IHJ002: Incident Respond Report

Date: 10 th June, 2026	Entry: IHJ002
Description	Handled a two-hour volumetric Denial of Service (DoS) attack, containing network congestion and hardening boundary firewalls using the NIST Cybersecurity Framework (CSF 2.0).
Tool(s) used	tcpdump, Wireshark, Splunk SIEM
The 5 W's (Who, What, where, when and why)	• Who: An organized, financially motivated cybercrime group. • What: Adversaries launched an unmitigated ICMP Echo Request flood that saturated network bandwidth, exhausted router state queues, and severed access to internal file servers, dev nodes, and communication tools. • Where: Corporate edge firewall and core enterprise routing backbone connecting internal departments and critical shared resources. • When: The breach began at 14:18:32 and caused a complete network outage for approximately two hours until services fully recovered at 16:05:32. • Why: The attack succeeded because the boundary firewall lacked basic ICMP rate-limiting thresholds and spoofed source IP verification, allowing the threat actor to overwhelm network buffers and disrupt normal business availability.
Additional notes	• The organization must never pay the ransom, as compliance does not guarantee data return, funds criminal infrastructure, and marks the firm as an easy target. Long-term prevention requires offline backups, regular software patching, multi-factor authentication, and workforce phishing awareness training.

Journal Entry IHJ003: Vulnerability Assessment Report

Date: 13 th July, 2026	Entry: IHJ003
Description	Deep packet capture analysis conducted to inspect raw network streams, trace protocol exchanges, and isolate anomalous or malicious network behavior.
Tool(s) used	Wireshark, tcpdump.
The 5 W's (Who, What, where, when and why)	• Who: Matthew Abegyah, Cybersecurity Analyst. • What: Protocol analysis and packet sniffing across captured .pcap files to identify unusual traffic spikes, connection states, and malformed frames. • Where: Border gateway interfaces and core corporate switching nodes. • When: Packet analysis session during the post-incident verification window. • Why: To diagnose low-level packet anomalies, detect active reconnaissance, and verify whether traffic matches authorized communication baselines.
Additional notes	Wireshark and tcpdump capture filters and stream-following features help analysts cut through high-volume noise to inspect suspicious transmission patterns and create precise Snort/firewall filtering rules.

Journal Entry IHJ004: Linux DAC & File Permissions Audit

Date: 27 th July, 2026	Entry: IHJ004
Description	Completed a filesystem access-control audit within an enterprise research workspace to identify and remediate Discretionary Access Control (DAC) permission drift using Linux command-line utilities.
Tool(s) used	Linux Bash Shell, POSIX Inode Utilities (ls -la, chmod).
The 5 W's (Who, What, where, when and why)	• Who: Matthew Abegyah (Lead Systems & Security Analyst) auditing multi-user workspace assets assigned to <i>researcher2</i> and <i>research_team</i>. • What: Audited POSIX file permission strings and corrected critical access violations, including removing global public write permissions (<i>-rw-rw-rw-</i>) from <i>project_k.txt</i>, revoking group traversal access (<i>drwx--x--</i>) on the <i>drafts/</i> directory, and fixing inconsistent write privileges on <i>.project_x.txt</i>. • Where: Enterprise Linux server environment at the local path <i>/home/researcher2/projects</i>. • When: Executed during a scheduled access-control security review. • Why: Unchecked default file creation masks and permission creep exposed critical research assets to unauthenticated local modifications, data corruption, and unauthorized directory traversal by non-owners.
Additional notes	Applied symbolic chmod commands (chmod o-w and chmod g-x) to restrict files and directories back to the Principle of Least Privilege (PoLP), locking the <i>drafts/</i> directory exclusively to the asset owner (<i>drwx-----</i>).

Journal Entry IHJ005: SQL for Security Telemetry Analysis (DONE)

Date: 17 th August, 2026	Entry: IHJ005
Description	Utilized relational database queries (SQL) to triage suspicious authentication anomalies across corporate login streams and segment organizational endpoint assets to accelerate security patch deployments.
Tool(s) used	MySQL Enterprise RDBMS and Structured Query Language (SQL).
The 5 W's (Who, What, where, when and why)	• Who: Matthew Abegyah (Lead Systems & Security Analyst) querying the <i>log_in_attempts</i> and <i>employees</i> database tables. • What: Executed targeted SQL queries using boolean logic (AND, OR, NOT) and pattern wildcards (LIKE '%') to isolate 19 after-hours failed login attempts, review authentication records across an active 48-hour intrusion window, filter out benign traffic from Mexico, and generate precise machine lists for endpoint patch updates. • Where: Corporate edge authentication gateways, perimeter logging databases, and internal directory tables across multiple physical office buildings. • When: Focused on post-business-hours activity (after 18:00). • Why: Threat actors conducted off-hours brute-force attempts to evade daytime detection, requiring fast database-level log analysis to trace malicious connections and isolate unpatched machines across specific business units without impacting already updated systems
Additional notes	Combining negative logic (<i>WHERE NOT department = 'Information Technology'</i>) with trailing wildcard operators (<i>office LIKE 'East%'</i>) yielded sub-second query performance while spotting inventory tracking issues such as unassigned devices (NULL device IDs) that require manual follow-up.

Journal Entry IHJ006: Automated Access Control With Python

Date: 1 st September, 2026	Entry: IHJ006
Description	Developed and tested a Python access-control algorithm to automate identity validation and hardware asset attribution, eliminating manual registry verification at physical security checkpoints
Tool(s) used	Python 3.x, Jupyter Notebook
The 5 W's (Who, What, where, when and why)	• Who: Matthew Abegyah (Security Analyst) verifying employees and their assigned hardware assets. • What: Designed an automated two-factor authorization script (<code>login()</code>) that evaluates username validity against approved directories and verifies. • Where: Corporate physical security desks, IT service workstations, and network provisioning checkpoints. • When: Completed during internal access-governance automation testing. • Why: Manual spreadsheet lookups cause queue delays, introduce human error, and allow unapproved personal devices (BYOD) or swapped hardware to connect to internal networks.
Additional notes	The algorithm enforces fail-closed logic: unrecognized usernames or mismatched device serials fail immediately and trigger security warning flags before network access is granted. Dynamic list operations (<code>.append()</code> and <code>.remove()</code>) ensure onboarding and offboarding keep user credentials and device IDs synchronized without orphaned records.